ChatGPT

Deep Research Report on ACGME and ABMS Competencies, Milestones, ABIM Competency Model, and an NLP-Based Behavioral Measurement Pipeline for Clinical Communication

Executive summary

The ABMS/ACGME “Six Core Competencies” (Medical Knowledge; Patient Care & Procedural Skills; Practice-Based Learning & Improvement; Systems-Based Practice; Interpersonal & Communication Skills; Professionalism) define the competency-based foundation that underlies U.S. graduate medical education and board certification standards. Official ABMS standards explicitly describe these competencies (including patient-centered decision-making within Patient Care and effective information exchange within Interpersonal & Communication Skills), and require boards to incorporate all six. [1]

ACGME Milestones operationalize those competency domains into specialty-specific, developmental “narrative descriptors” (Levels 1–5) intended primarily for formative assessment, shared language, and program improvement—not as externally imposed high-stakes decision tools. ACGME has further cautioned that individual Milestones data were not designed for external high-stakes decisions and that such use can distort ratings. [2]

For clinical communication behaviors, practical rubrics like the Calgary–Cambridge Guide (consultation structure and micro-skills), NURSE (empathic response micro-skills), and AHRQ’s SHARE approach (shared decision-making steps) can be mapped to core competencies—especially Interpersonal & Communication Skills and Professionalism—and to specific Milestones subcompetencies such as Internal Medicine ICS1 (Patient- and Family-Centered Communication), which explicitly includes shared decision-making and mitigation of communication barriers at higher developmental levels. [3]

Your described experiment—pivoting from a synthetic “chat note / NoteChat-style” dataset to real clinician–patient conversations to strengthen ground truth; using LLM-based labeling supported by rubric-grounded prompting and a small set of exemplar conversations per construct; complemented by NER-based labeling (most plausibly de-identification and/or entity-focused preprocessing); and then training downstream models to produce “final grades” aligned to Calgary and NURSE constructs—fits a well-supported direction in the literature: (a) synthetic conversations are useful for bootstrapping but can diverge from real-world distributions; (b) LLMs can serve as scalable annotators but require careful validity checks, bias monitoring, and human-in-loop governance; and (c) “NLP-screened human abstraction” is a proven approach to reduce manual effort while protecting measurement validity in high-stakes contexts. [4]

The most defensible path for stakeholders is to integrate NLP-derived metrics as *decision support* within a competency assessment ecosystem: validate against human-coded gold sets and Milestones-aligned rubrics; calibrate uncertainty and set conservative routing thresholds; audit fairness by subgroup and

context; apply PHI/consent safeguards; and keep final judgments with trained reviewers (especially if results could affect advancement/certification). [5]

Sources accessed and scope notes

Connectors accessed (enabled): GitHub, SharePoint, Monday.com (searched). Monday.com searches did not return relevant artifacts for the queried terms ("ChatNote," "NURSE"). (No Monday.com sources were ultimately used as evidence.)

Internal GitHub repository accessed (restricted to your selection): **ABIM-ETD/competency-map**. Files retrieved via connector: - `README.md` (project overview, data import pipeline references) — GitHub UI view.
- `config/import_map.yaml` (Excel-to-database column mappings; references the master Milestones workbook used by the app).

Internal SharePoint artifacts located and/or retrieved (file names and SharePoint paths are verifiable from search results below): - **MASTER Milestones 2 with Definitions and Harmonized 20250818.xlsx** (appears in the Competency Model folder and also in the AI Behavior Analysis research folder).
- **Labelling_BehaviorAnalysis.docx** (in ETD-RF AI Behavior Analysis research folder; includes the Calgary/NURSE/SHARE label list and mentions ABMS/ACGME alignment fields).
- **Project summary.docx** (personal SharePoint results; describes a "ChatNote-style" dataset plan and a Label Studio annotation workflow).

Important scope limitation: The internal documents retrieved via SharePoint search in this session clearly support the *label ontology and original plan* (synthetic HuggingFace "ChatNote-style" dataset + manual Label Studio). They did **not** explicitly document the *newer pipeline details* you described (pivot to real transcripts, LLM labeling for 985 conversations, exemplar selection of 8–10 conversations per construct, Qwen/Llama model selection). Those newer steps are therefore treated as **project-team described implementation details** in this report; the "deep research" contribution here is to connect them to external evidence and best practices, and to align them to the competency/milestones frameworks.

Core competency frameworks and Milestones

ABMS standards for Initial Certification require ABMS Member Boards to incorporate **all six ABMS/ACGME Core Competencies**, and the standards include "annotation" text that defines each competency domain (including explicit references to patient-centered decision-making, effective listening, nonverbal communication, health literacy, ethics, and humanism). [1]

ACGME Milestones were developed to create a specialty-specific developmental continuum and a "common language" for progression in competence; ACGME-I's overview emphasizes that Milestones provide narrative descriptors along a continuum and should be used for formative purposes, informing (not replacing) assessment tools aligned to actual learner tasks. [6]

ACGME also states that the Milestones "were not designed or intended" for external high-stakes decisions (e.g., licensing/credentialing), warning of negative incentives and potential inflation if misused. This is directly relevant when considering automated measurement outputs—NLP grades should be positioned as

formative or *decision-support* unless validated for a specific high-stakes context under rigorous governance. [7]

Competencies and where clinical communication "lives" in the framework

Clinical communication behaviors (rapport, agenda-setting, empathy, shared decision-making, teach-back, closing/safety-netting) are most directly aligned to **Interpersonal & Communication Skills** and often to **Professionalism** (humanism, sensitivity, respectful partnerships), with downstream impact on **Patient Care & Procedural Skills** (patient-centered decisions, adherence, safety). ABMS's official competency descriptions explicitly link these ideas to the relevant competencies. [8]

Milestone structure example: Internal Medicine ICS1 explicitly includes shared decision-making

The Internal Medicine Milestones (v2.1) provide a clear example of how a competency becomes a developmental scale. In **ICS1 (Patient- and Family-Centered Communication)**, Level 4 includes "shared decision making" and "mitigates communication barriers," and Level 5 focuses on coaching others—language that creates a natural bridge from observable conversation behaviors to milestone judgments. [9]

A more behavior-granular version of the same idea appears in the ACGME-I Internal Medicine Milestones (v2.0), where descriptors include setting the agenda, verifying understanding, eliciting values/goals/preferences, reflecting on bias, and using shared decision-making to align preferences to options—elements that can be linked to Calgary/NURSE/SHARE labels in transcript-based modeling. [10]

ABIM's role and Milestones context

ABIM's Milestones pages describe Milestones as outcomes-based expectations within the six domains, co-convened by ACGME/ABIM/AAIM working groups, used for normative data and program improvement, and to demonstrate accountability of GME's effectiveness. [11]

Communication behavior frameworks and mapping to competencies and milestones

Evidence-based behavior frameworks relevant to transcript labeling

Calgary–Cambridge is widely used as a consultation structure and communication micro-skills framework; its referenced observation guides were designed to delineate and structure skills that aid doctor–patient communication. [12]

NURSE (Name, Understand, Respect, Support, Explore) is commonly taught as a structured approach to empathic responses, with practical training materials illustrating how each step is operationalized as language in clinical interactions. [13]

AHRQ's SHARE approach is a formal model for shared decision-making that defines five steps (Seek participation; Help explore/compare options; Assess values/preferences; Reach a decision; Evaluate), designed for clinician training and implementation. [14]

Mapping table: example observable behaviors to competencies and Internal Medicine ICS1 milestones

The table below illustrates how transcript-visible behaviors (including your Calgary/NURSE constructs) can be mapped upward to competency domains and Milestones language. The Milestones citations here use official Internal Medicine Milestones language (US v2.1 and ACGME-I v2.0), because these documents contain explicitly behavior-linked descriptors (shared decision-making; mitigating communication barriers; agenda-setting; eliciting values/goals). [15]

Observable behavior in transcripts	Framework tag in your label space	Primary ABMS/ ACGME competency domain	Internal Medicine ICS1 Milestones anchor (illustrative)
Sets agenda at start ("What would you like to focus on today?"; confirms list)	Calgary: `cc_agenda_set`	Interpersonal & Communication Skills [16]	ACGME-I ICS1 includes "setting the agenda" and "verifying understanding" in developmental descriptors [10]
Uses respectful language / rapport-building nonverbal cues (as reflected in text: respectful address, validation)	Calgary: `cc_opening`	Interpersonal & Communication Skills; Professionalism [16]	US IM ICS1 Level 1 emphasizes respect/rapport [9]
Names emotion ("It sounds like you're worried.")	NURSE: `nurse_name`	Professionalism; Interpersonal & Communication Skills [16]	Supports therapeutic relationship behaviors; contributes to higher-level shared decision-making by addressing emotion barriers [17]

Observable behavior in transcripts	Framework tag in your label space	Primary ABMS/ ACGME competency domain	Internal Medicine ICS1 Milestones anchor (illustrative)
Shows understanding / respectful acknowledgment (“That makes sense.” / “You’ve been working hard.”)	NURSE: `nurse_understand`, `nurse_respect`	Professionalism; Interpersonal & Communication Skills [16]	Aligns with “establishes and maintains a therapeutic relationship” (ICS1 L2–L3 concepts) [9]
Offers support / partnership (“I’ll be with you through this.”)	NURSE: `nurse_support`	Professionalism; Interpersonal & Communication Skills [16]	Supports barrier mitigation and trust building needed for complex SDM [18]
Elicits values/ preferences; explores goals (“What matters most to you?”)	SHARE: `sh_assess_values_preferences`	Patient Care (patient-centered decision-making); Interpersonal & Communication Skills [16]	ACGME-I ICS1 explicitly includes eliciting values/ goals/preferences [10]
Compares options with risks/benefits in dialogue	SHARE: `sh_help_compare_options`	Patient Care; Interpersonal & Communication Skills [16]	US IM ICS1 Level 4: “shared decision making” regardless of complexity [19]
Summarizes and checks back (teach-back-adjacent)	Calgary: `cc_summary_checkback`	Interpersonal & Communication Skills [16]	ACGME-I ICS1 includes verifying understanding and aligning plans [10]
Closes with next steps and safety-netting	Calgary: `cc_closing_next_steps`	Patient Care; Interpersonal & Communication Skills [16]	Contributes to patient-centered plan and barrier mitigation; often reflected in ICS1 higher-level proficiency [9]

Internal note: your SharePoint labeling doc indicates this combined label space (Calgary, NURSE, SHARE) and mentions ABMS/ACGME alignment fields, consistent with the mapping approach above.

Data and labeling pipeline deep dive with supporting research connections

This section connects your described pipeline to the best available evidence on (a) synthetic vs real transcripts, (b) LLM-assisted labeling with rubric-grounded prompts, and (c) NER-driven labeling steps (most commonly PHI de-identification in healthcare transcripts).

Synthetic "chat note / NoteChat-style" data as a bootstrap, and why pivoting to real conversations improves ground truth

Your internal project summary references a plan to use a HuggingFace "ChatNote-style" doctor–patient dataset to label behaviors and train models. Externally, there are well-described synthetic clinical dialogue datasets—e.g., **NoteChat**, which generates synthetic doctor–patient conversations conditioned on clinical notes using LLM-based roleplay modules, releasing a large dataset intended to overcome limited availability of real conversations. [20]

However, both general ML and healthcare NLP experience suggest a strong reason to pivot toward real transcripts when the objective is *valid measurement of human behaviors*: model performance and rubric interpretation can be sensitive to distribution shifts between synthetic and authentic interactions. The NoteChat ecosystem itself exists because real conversations are limited and constrained, implicitly acknowledging that synthetic data is a workaround rather than a perfect substitute. [21]

More broadly, research on "model collapse" and synthetic-data recursion highlights a foundational risk: over-reliance on generated data can reduce representation of the "tails" of real distributions. While that work is framed around generative model training, the core implication—**genuine human interaction data becomes increasingly valuable**—supports your decision to pivot toward real doctor–patient conversations for stronger ground truth. [22]

LLM-based labeling at scale: why rubric-grounded, few-shot prompting is a reasonable strategy

Your described method—selecting ~8–10 exemplar conversations per construct, crafting prompts that encode the rubric, and using an LLM to label a much larger set—is a practical instantiation of **in-context learning / few-shot prompting**, where task instructions + demonstrations guide model behavior. This is the central paradigm introduced in the "few-shot learners" literature. [23]

Empirically, research indicates LLMs can be surprisingly effective as annotators, sometimes matching or exceeding crowd-worker performance on text labeling tasks, including with higher agreement and lower cost (noting domain differences). [24] More recent syntheses (surveys) organize LLM-based annotation into: generating annotations, assessing annotation quality, and "learning with" LLM-generated labels—exactly the three-phase program your pipeline implies (label → validate → train downstream model). [25]

At the same time, the literature also flags limits: LLM annotation agreement with humans can be moderate and can reproduce biases present in training data; this reinforces the need for human-in-loop review, bias audits, and calibration. [26]

NER in transcript pipelines: de-identification and structured extraction as enablers

When transcripts involve real clinical conversations, **PHI risk** becomes a gating constraint. A well-established approach is NER-style tagging of PHI entities followed by replacement/redaction. The i2b2/UTHealth 2014 de-identification corpus paper details a rigorous PHI annotation and surrogate generation process under HIPAA guidelines, using double annotation + adjudication and reporting inter-annotator agreement/quality metrics. [27]

If your "NER method to label the data" includes PHI tagging/redaction (a common necessity when moving from synthetic to real conversations), the i2b2 methodology provides a defensible precedent for why NER-based labeling is part of "doing it right" before scaling LLM labeling or model training. [27]

Why "top N excerpts" and "few demonstrations per construct" is consistent with evidence from adjacent clinical NLP measurement tasks

A closely analogous pattern appears in published work on automating evaluation of physician communication: in prostate cancer consultations, researchers developed NLP models to identify high-yield sentences related to key concepts and found that grading quality using only the **top 10 sentences** achieved a feasibility/accuracy balance. [28] This aligns with your approach of (a) using limited exemplar segments to anchor the rubric and (b) scaling scoring/labeling across many transcripts.

Another strong precedent for *hybrid pipelines* is NLP as an efficient screen plus targeted human abstraction: in measuring goals-of-care discussions as a trial outcome, NLP-assisted abstraction drastically reduced manual hours while maintaining strong sensitivity and managing power loss due to misclassification. [29] This supports your strategy of using automation to reduce human burden while maintaining governance and validity.

Generalizability beyond healthcare: why the method transfers to transcript-heavy industries

Even though the constructs (Calgary/NURSE) are clinical, the *measurement problem* is cross-industry: mapping unstructured dialogue to rubric-defined competencies/behaviors. Industries with dense transcripts include: - legal depositions/hearings (argument quality, politeness, clarity, responsiveness), - customer support/call centers (empathy, resolution behaviors, compliance scripts), - education/advising (feedback quality, scaffolding, inclusion), - therapy/coaching (reflective listening patterns—highly sensitive contexts).

The research base on LLMs as annotators and on transcript-based NLP measurement is not inherently clinical; it generalizes to any domain where (1) behaviors are defined by a rubric, (2) human measurement is expensive/variable, and (3) outcomes depend on trustworthy operationalization of constructs. [30]

Assessment and measurement challenges in "behavioral" competency measurement

Manual measurement of communication and behavioral skills is difficult because constructs like empathy, rapport, and shared decision-making are partially subjective, context-dependent, and can be expressed in

many linguistic forms. A systematic review of physician–patient communication measures emphasizes that the validity of findings depends heavily on measurement quality and psychometrics, and that many instruments vary in methodological quality. [31]

Reviews of communication competence tools similarly highlight that many instruments exist but psychometric properties and practicality vary, and that high-quality assessment is resource-intensive. [32]

These realities map directly onto what you are solving: scaling measurement of *behaviors* in transcripts is hard because manual coding is slow, expensive, and can have limited reliability—especially as you move from a pilot to hundreds or thousands of encounters.

For competency frameworks, this measurement difficulty matters because Milestones and competencies are designed as professional-development tools; if measurement becomes overly automated without careful safeguards, you risk (a) false precision, (b) distortions from high-stakes coupling, and (c) fairness harms. ACGME's caution against external high-stakes use of individual Milestones data is a strong governance signal for any automated "grading" approach: it should be framed as formative/feedback unless validated and governed accordingly. [7]

Practical recommendations for integrating NLP-derived metrics into competency assessment

The recommendations below assume your current workflow: real+simulated transcripts (~985), rubric-based constructs (Calgary + NURSE), LLM labeling (Qwen/Llama variants), and downstream model training to predict "final grades." (Where numbers/models appear, they reflect your described implementation; the validation steps apply regardless of exact counts.)

Validation and governance steps aligned to Milestones and competency use

Define construct validity explicitly (rubric-to-milestone mapping). Start with a written mapping from each construct label (e.g., `cc_agenda_set`, `nurse_respect`) to (a) ABMS/ACGME competency domain(s) and (b) the specific Milestone subcompetency descriptors you intend to support (e.g., IM ICS1 Level 4 shared decision-making). Use official Milestone language as the "anchor text." [33]

Create a stratified, human-coded gold set (even small) across specialties, encounter types, and difficulty. Because communication behaviors are sparse/imbalanced (e.g., explicit empathy statements may be rare), oversample likely-positive contexts to get stable per-class precision/recall estimates.

Use an "NLP-screened human review" pattern for higher-stakes outputs. The JAMA Network Open study shows that combining NLP screening with targeted human abstraction can preserve sensitivity while massively reducing labor. Use this for: low-confidence cases, high-impact decisions, or fairness-flagged slices. [29]

Do not position automated rubric grades as "Milestones ratings." Treat them as *evidence signals* that may inform faculty CCC discussions, coaching, or remediation plans. This aligns with ACGME's stated intent and avoids the pitfalls of turning Milestones into externally high-stakes metrics. [34]

Recommended metrics for your labeling model and downstream "grading" model

For each construct (Calgary and NURSE), report: - **Per-class Precision / Recall / F1** (because norms and prevalence differ by construct; macro averages can hide failure on rare classes).
- **Macro-F1** (primary headline for imbalance robustness).
- **Confusion matrices** (especially for ordinal Calgary scores if you keep 0–2).
- **Inter-annotator agreement (IAA)** between (a) LLM labels and (b) humans on a subset; for multi-label constructs consider agreement per label and overall. The i2b2 de-identification corpus illustrates how annotation quality can be quantified and audited using agreement metrics and adjudication processes. [27]
- **Calibration metrics** (e.g., Expected Calibration Error, Brier score) so that probability thresholds reflect real risk; calibration is critical for routing logic in human-in-loop workflows. The clinical-trial outcome paper underscores that misclassification affects downstream power and decisions; calibration and thresholding are therefore not "nice-to-have." [29]
- **Slice-based performance and fairness checks** (by site, speaker role distribution, patient demographics if available/allowed, language proficiency, encounter type). LLMs can replicate biases present in data, so monitoring for systematic differences is needed. [26]

Thresholding, human-in-loop design, and change management

Adopt a **tiered threshold policy**: - *Auto-accept* only when model confidence is high and the construct is low-risk (e.g., detecting a clear "agenda set" phrase).
- *Human review required* for low-confidence predictions, high-stakes contexts, or constructs with known subjectivity (empathy nuance).
- *Escalation* when fairness checks breach pre-set gaps (e.g., >X% absolute drop in F1 for a demographic slice), similar to "fairness-as-tests" practices.

This approach mirrors both ACGME's formative framing and empirical evidence that hybrid workflows can preserve quality while scaling. [35]

Privacy and PHI handling when using real transcripts

Before any labeling at scale, ensure a defensible PHI risk posture: - Use NER-based PHI de-identification and/or surrogate replacement pipelines; the i2b2 2014 methodology provides a blueprint (double annotation, adjudication, and multiple checks) for high-confidence de-identification in clinical text. [27]
- Maintain strict access controls, audit logs, and clearly documented data flows.

Suggested visualizations

Mermaid timeline for Milestones progression (using IM ICS1 as an example): [9]

```
timeline
  title IM ICS1 (Patient- and Family-Centered Communication) — developmental
continuum
  Level 1 : Respect/rapport basics
  Level 2 : Therapeutic relationship in straightforward encounters
  Level 3 : Therapeutic relationship in challenging encounters; identifies
```

```
complex barriers
  Level 4 : Shared decision-making regardless of complexity; mitigates barriers
  Level 5 : Coaches others; models barrier mitigation
```

Mermaid flowchart for a validation and deployment pipeline aligned to your rubric-based labeling approach: [36]

```
flowchart TD
  A[Define constructs + rubric\n(Calgary/NURSE) + milestone anchors] -->
B[Collect transcripts\n(real + simulated)]
  B --> C[NER-based preprocessing\n(PHI de-id + structure tags)]
  C --> D[Curate 8–10 exemplar conversations\nper construct + prompt]
  D --> E[LLM labeling at scale\n(Qwen/Llama best-of per construct)]
  E --> F[Gold set human annotation\n(stratified sample)]
  F --> G[Evaluate label quality\nper-class F1 + IAA + bias slices]
  G --> H[Train downstream scoring model\n(multilabel/ordinal)]
  H --> I[Calibrate probabilities\nand set thresholds]
  I --> J[Deploy decision support\nwith routing: auto/queue/escalate]
  J --> K[Ongoing monitoring\ndrift + fairness + periodic revalidation]
```

Concrete recommendations tailored to your described workflow

Use your current **"985 × construct" labeled datasets** as *training inputs*, but treat **LLM labels as silver-grade** unless validated against an adjudicated human gold set. This aligns with both the promise (LLMs can annotate effectively) and the limitations (bias/partial agreement) documented in the LLM-annotation literature. [37]

Adopt **construct-by-construct model selection** (your "Qwen or Llama depending on which performed well") but require the selection to be justified by *per-construct* validation metrics, not only overall accuracy: rare constructs like certain NURSE behaviors can fail silently under aggregate metrics. [38]

If you intend to translate construct detection into a **single "final grade"** for a clinician, add a transparent aggregation model (e.g., weighted rubric score with rationale) and validate whether that composite correlates with external criteria (OSCE ratings, OPTION5/SDM tools, CCC judgments). This mirrors how published systems validated "quality evaluation" using targeted sentence selection and human-coded quality hierarchies in real consultations. [39]

Finally, position the system as **scalable feedback infrastructure** rather than a replacement for competency committees. This is consistent with ABMS/ACGME competency intent, ACGME's Milestones governance cautions, and the empirical success of hybrid NLP + human abstraction models. [40]

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[1] [8] [16] [40] https://www.abms.org/board-certification/board-certification-standards/standards-for-initial-certification/

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